

Nonnegative Matrix Factorization Common Spatial Pattern in Brain Machine Interface

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Abstract— Fast and accurate discrimination of Electroencephalography (EEG) data is necessary for controlling brain machine interface. This paper introduces a novel method to discriminate 2-class motor imagery states (left and right hand) using nonnegative matrix factorization (NMF), common spatial pattern (CSP) and random forest. Conventionally CSP is used after extracting frequency band segment of EEG signal, which is called bandpass-filtered CSP (BPCSP). Especially filter bank CSP (FBCSP) has been extensively used to extract feature vectors from EEG data. However in these methods, the range of frequency band needed to be specified in advance and the performance depends on the selected frequency band. Our new method can decide the frequency band automatically by using NMF (NMFCSP). After the feature vectors were extracted from EEG data, random forests (RF) method was adopted as a classification algorithm. The mean accuracy rate of 2-class classifier using NMFCSP was $78.8 \pm 3.27\%$. This is higher than the accuracy rate of BPCSP ($64.4 \pm 8.53\%$) and FBCSP ($68.4 \pm 6.81\%$).

Keywords—EEG classification; motor imagery; common spatial pattern; nonnegative matrix factorization; random forest;

Introduction

Our purpose is to develop non-invasive brain machine interface (BMI). Electroencephalography (EEG) is the most applicable tool for non-invasive BMI systems due to its high temporal resolution and portability. However EEG doesn't have high spatial resolution and signal-to-noise ratio. Therefore we have to reduce artifacts and extract useful feature vectors. The common spatial pattern (CSP) is an effective algorithm to analyze spatial patterns of 2-class motor imagery in the EEG based BMI. Conventionally CSP is used after reducing artifact by bandpass filter (BPCSP). Especially filter bank CSP (FBCSP) is known as a superb method [1]. FBCSP is an algorithm which selects the best pattern of frequency band segments and analyzes spatial patterns of the frequency band segments by using CSP. However the range of frequency band should be predefined and the performance depends on the frequency band. To address this problem, we used nonnegative matrix factorization (NMF). NMF is one of the multivariate analysis methods to find basis vectors which construct high dimensional nonnegative data. The basis vectors denote features of them, therefore NMF can summarize resemble features in a single component and express the high dimensional data with less feature vectors. We adopted NMF to calculate the best patterns of frequency band segments

automatically and calculate the accuracy rate of 2-class motor imagery states (left and right hand).

I. PROCEDURE

The procedure of NMFCSP is illustrated in Fig. 1. It has six steps of EEG signal processing. First, continuous wavelet transform with fast Fourier transform algorithm (CWT with FFT) is applied to multi-channel EEG signals x (the number of rows of x equals to the number of EEG channels (CH) and that of columns of x equals to time point (TIME) $\times$ the number of trials (TRIAL) $\times$ the number of classes (CLASS)) to construct nonnegative matrix data X (the number of rows of X equals to the number of frequency components (FQ) $\times$ CH and that of columns of X equals to TIME $\times$ TRIAL $\times$ CLASS) which correspond to the time-frequency distribution of the respective signals. We set the frequency band for feature extraction from 8 to 25Hz (α to β bands), which is frequently used in motor imagery analysis. Second, NMF decomposes X into two matrices, the basis matrix A (the number of rows of A equals to FQ $\times$ CH and that of columns of X equals to the number of basis vectors (k)) and the encoding matrix S (the number of rows of S equals to k and that of columns of S equals to TIME $\times$ TRIAL $\times$ CLASS). The k is a parameter we have to decide. In this study we used k values ranged from 3 to 30. Third, we reconstruct the time-frequency matrices $X'i$ ($i = 1, 2, \dots, k$) by multiplying respective basis vectors and encoding vectors. Forth, inverse Cwt with FFT is applied to respective $X'i$ (the number of rows of $X'i$ equals to FQ $\times$ CH and that of columns of $X'i$ equals to TIME $\times$ TRIAL $\times$ CLASS) to reconstruct signal processed EEG data $x'i$ (the number of rows of $x'i$ equals to CH and that of columns of $x'i$ equals to TIME $\times$ TRIAL $\times$ CLASS). Fifth, CSP is used to analyze spatial patterns of 2-class motor imagery and accuracy rates are calculated by using random forests (RF) and leave-one-out cross-validation (LOO-CV) on each of $x'i$. Finally, we select several $x'i$ which has high accuracy rates and discriminate their $x'i$ by using RF and LOO-CV. We found the best combination of $x'i$ by extracting it in descending order of accuracy rate and calculating accuracy rate again with the extracted pattern.

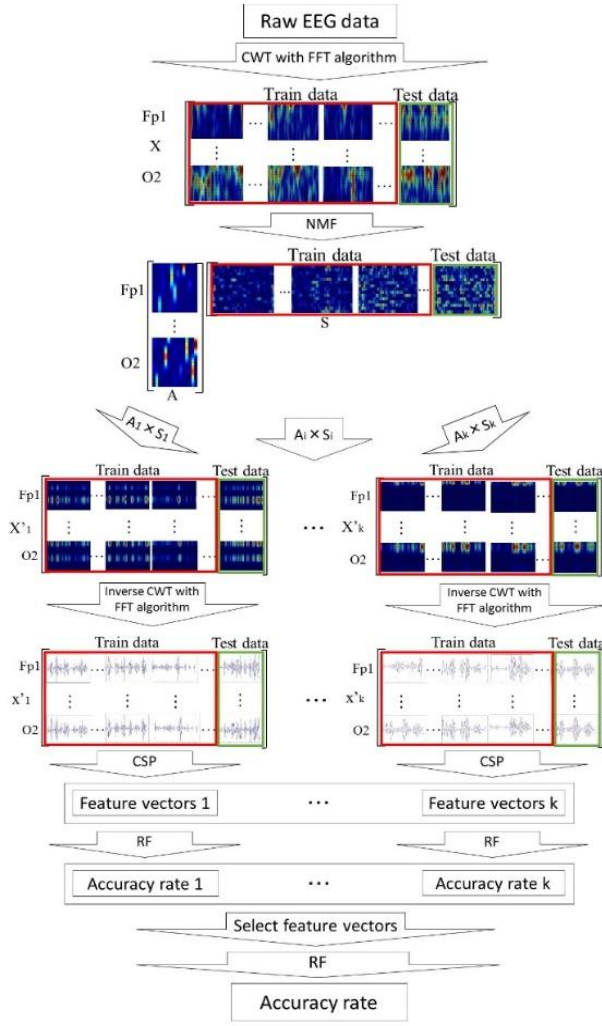


Fig. 1 The procedure of NMFCSP algorithm.

II. ALGORITHM

A. Continuous Wavelet Transform with Fast Fourier Transform Algorithm

CWT calculates changes in the amplitude of the signal at an arbitrary frequency, which is defined by the following:

$$W(a, b) = \int_{-\infty}^{+\infty} \frac{1}{\sqrt{|a|}} \overline{\Psi\left(\frac{t-b}{a}\right)} s(t) dt \quad (1)$$

where $s(t)$ denotes a signal and the $\Psi\left(\frac{t-b}{a}\right)$ denotes a basis wavelet. The dilatation and the temporal shift of the Ψ is represented by factors a and b , respectively. CWT compares a signal s and a basis wavelet Ψ with changing the scale a and the shifting b . $W(a, b)$ shows correlation of the high frequency when the scale a is small and it shows correlation of low frequency when the scale a is large. There are many kinds of basis wavelet Ψ such as Morlet wavelet and Meyer wavelet. We used Morlet wavelet as a basis wavelet which is frequently used in EEG analysis [2] [3].

$W(a, b)$ includes negative values because it is correlation between a signal s and a basis wavelet Ψ . Since NMF accepts only positive values to be analyzed, we used CWT with FFT algorithm which is defined by the following:

$$W(a, b) = \frac{1}{\sqrt{|a|}} F^{-1}[F[\overline{\Psi}] * F[x]] \quad (2)$$

where $F[\cdot]$ denotes FFT, $F^{-1}[\cdot]$ denotes inverse FFT and $*$ denotes convolution. The detail of FFT algorithm is found in [4]. The elements of $W(a, b)$ which are calculated by this algorithm have complex numbers. We therefore take the absolute value $|W(a, b)|$ which is nonnegative for the further analysis. We can reconstruct the complex elements from $|W(a, b)|$ by calculating the angles of $W(a, b)$. This is useful when we reconstruct signal processed EEG data by using inverse CWT with FFT.

B. Nonnegative Matrix Factorization

NMF is one of multivariate analysis methods for nonnegative data such as image files and time-frequency spectrograms. NMF decomposes a nonnegative matrix $X \in R^{m \times n}$ (m and n are the number of rows and columns of X , respectively) into two nonnegative matrices, the basis matrix $A \in R^{m \times k}$ and the encoding matrix $S \in R^{k \times n}$ (k is the predetermined number of basis vectors). The result of the NMF shows several patterns which compose a nonnegative matrix X and they are sparse. The conventional approach to find A and S is to minimize the difference between X and AS as the followings:

$$\min D(X||AS) \quad (3)$$

$$\text{subject to } A_{ic} \geq 0, S_{cj} \geq 0 \quad (4)$$

where $D(X||AS)$ shows distance between X and AS and c denotes the element of basis vector. We use Kullback-Leibler (KL) divergence as a distance because it facilitates more accurate matching and selection of computed endmembers [5]. KL divergence is defined by the following:

$$D_{KL}(X||AS) = \sum_{i,j} (-X_{ij} \log \frac{X_{ij}}{\sum_k A_{ik} S_{kj}} - X_{ij} + \sum_c A_{ic} S_{cj}) \quad (5)$$

where D_{KL} denotes KL divergence. It is known that the following update rules find A and S which minimize $D_{KL}(X||AS)$ [6] [7]:

$$A_{ic} \leftarrow A_{ic} \frac{\sum_j \frac{X_{ij}}{(AS)_{ij}} S_{cj}}{\sum_j S_{cj}}, S_{cj} \leftarrow S_{cj} \frac{\sum_i \frac{X_{ij}}{(AS)_{ij}} A_{ic}}{\sum_i A_{ic}} \quad (6)$$

C. Common Spatial Pattern

Common Spatial Pattern is a two-class classifier of multichannel data. CSP filter maximizes the ratio of the variance of the data in one class while minimizes that in another class. Band-pass filtered EEG-data is defined by the following:

$$E_j \in \mathbb{R}^{m \times k} \quad (7)$$

Where j denotes the number of trials, m denotes the number of channels, and k denotes the number of data from each trial. We can estimate of the covariance matrix of E_j by

$$\Sigma^C = \sum_{j \in C^i} \frac{E_j E_j^T}{\text{trace}(E_j E_j^T)}, \quad (i \in [1, 2]) \quad (8)$$

CSP filter W is given by simultaneous diagonalization between Σ^1 and Σ^2 :

$$W^T \Sigma^{(1)} W = \Gamma^{(1)}, W^T \Sigma^{(2)} W = \Gamma^{(2)} \quad (9)$$

$$\Gamma^{(1)} + \Gamma^{(2)} = I \quad (10)$$

where $\Gamma^{(1)}$ and $\Gamma^{(2)}$ are diagonal matlix. CSP filter W coressponds to eigenvectors

$$W = [\omega_1, \dots, \omega_m] \quad (11)$$

The eigenvectors is given by solving the genelized eigenvalue problem

$$\Sigma^{(1)} \omega = \lambda \Sigma^{(2)} \omega \quad (12)$$

D. Random Forests

RF classifier is one of ensemble learning algorithms which is consisted of many binary decision trees. A binary decision tree, which is illustrated in Fig. 2, has some nodes and leaves which output final result. Node has split function which decides one of the two branches to proceed. Split function is defined by the following:

$$h(f, \theta_k) \in (0, 1) \quad (13)$$

where f denotes feature vector that reached to the node. θ_k denotes parameter of split functions and k is the number of nodes. When feature vector is given to the tree, it reaches to the first node. At the node, a split function determines the appropriate branch to proceed. Repeating these decisions at

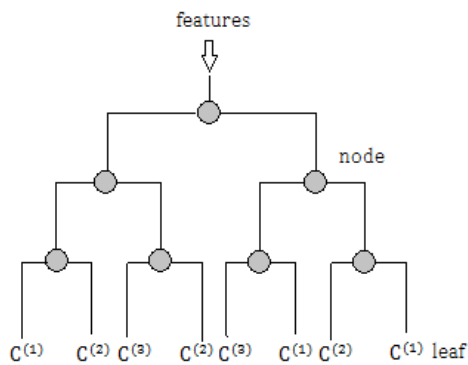


Fig. 2 Abinary decision tree

each node, the feature finally reaches to one of the leaves and output decision is made. One decision tree gives single classification result. Although a single decision tree does not always classify correctly, it is possible to compensate for the decision by combining the decisions from many trees. Features given to the trees are different depending on the tree. $f_m \in f$ shows an input to the m -th tree. RF classifier decides class by majority vote of many decision trees.

III. EXPERIMENTS

Eight healthy young subjects (20-25 years old) participated in the study (subject A, B, C, D, E, F, G and H). We measured their EEG signals from 15 electrodes distributed over the scalp (Fp1, Fp2, F3, Fz, F4, T3, C3, Cz, C4, T4, P3, Pz, P4, O1, O2), sampled at 125 Hz. The procedure of the experiment is illustrated in Fig. 3. Subjects gazed at a computer screen in front of them during the whole experiment. The screen alternately showed an instruction (4s) and a fixation point (3s) for 40 times. An instruction of single state among two states (left hand or right hand movement imagery) was randomly displayed at the beginning of each trial for 1s, followed by the 3s of count-down signs. After the count-down subjects performed the instructed state for 3s. To reduce artifacts, subjects were instructed not to blink and move his eyes. Subject performed two kinds of states each for 20 times.

IV. RESULTS & DISCUSSION

The result of analysis of 2-class motor imagery classifier using NMFCS, BPCSP, and FBCSP is shown in Table. 1. This indicates that the accuracy rate obtained from NMFCS is higher than that from BPCSP or FBCSP. FBCSP is an algorithm which selects the best pattern of frequency band fragments after dividing it. However we must decide the range of them beforehand. On the other hand NMFCS objectively decides the

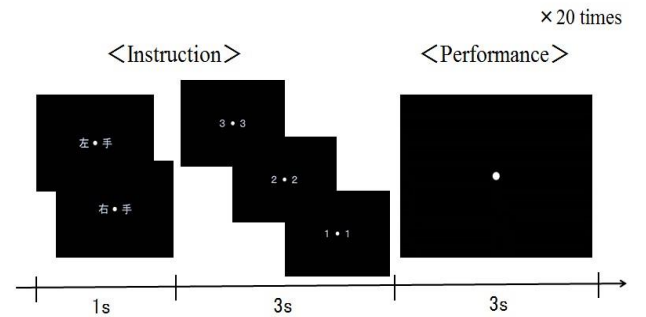


Fig. 3 The procedure of experiment. Instruction was first displayed to the subjects in Japanese. “左手” equals to the left hand and “右手” equals to the right hand. Subjects were encouraged to gaze at white fixation point in order not to move their eyes during experiment.

Table. 1 The accuracy rates of LOO-CV performed using BPCSP, FBCSP and NMFCSP. The parameter k on this table is the number of basis vectors.

Subject	BPCSP		FBCSP		NMFCSP	
	Accuracy	Frequency	Accuracy	Frequency	Accuracy	k
A	62.5%	12~25Hz	62.5%	18~20Hz	77.5%	k=29, k=30
B	72.5%	8~25Hz	70%	10~14Hz, 16~18Hz	82.5%	k=28
C	67.5%	12~20Hz	77.5%	12~16Hz, 18~22Hz	77.5%	k=21, k=24, k=30
D	62.5%	12~16Hz	60.0%	8~10Hz, 20~22Hz	80%	k=22
E	75%	12~25Hz	72.5%	12~14Hz, 16~18Hz, 22~25Hz	80%	k=27
F	47.5%	8~12Hz	60.0%	10~12Hz, 22~24Hz	72.5%	k=19, k=29
G	60%	16~20Hz	75%	10~18Hz	82.5%	k=20, k=26
H	67.5%	8~20Hz	70%	16~18Hz	77.5%	k=27, k=29
Ave	64.4%		68.4%		78.8%	

range of frequency band fragments and the pattern of the selected frequency range can be different channel to channel (Fig. 4). This is more suitable than subjectively determine the frequency band because NMF combines resemble features together. Moreover the weight of the frequency bands on each electrode can be different (Fig. 4). Measured brain activities can be different depending on the electrode positions, suggesting that the characteristic of NMFCSP is more effective to extract useful features than BPCSP or FBCSP.

V. CONCLUSION

We introduced a new method to automatically extract better feature vectors of 2-class motor imagery EEG data to overcome the conventional methods that subjectively decide feature vectors (frequency bands). Our proposed method achieved higher accuracy rate compared to the conventional methods. However we still have to decide the number of basis vectors which may affect the accuracy rate. In this paper we tried to change the basis from 3 to 30 therefore it took long time to find the best parameter. Further research is required to automatically determine the best number of basis vectors.

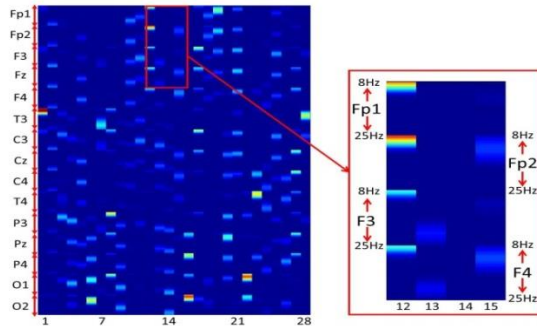


Fig. 4 The basis matrix calculated by the data of subject B. The number of basis vectors is 28 and frequency range is from 8 Hz to 25 Hz. On each sub-column corresponding to each electrode, the upper part indicates the low frequency component while the lower part indicates the high frequency component.

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